

0 6 . 1

Explain a property of iron ions that enables these ions to carry out their role in red blood cells.

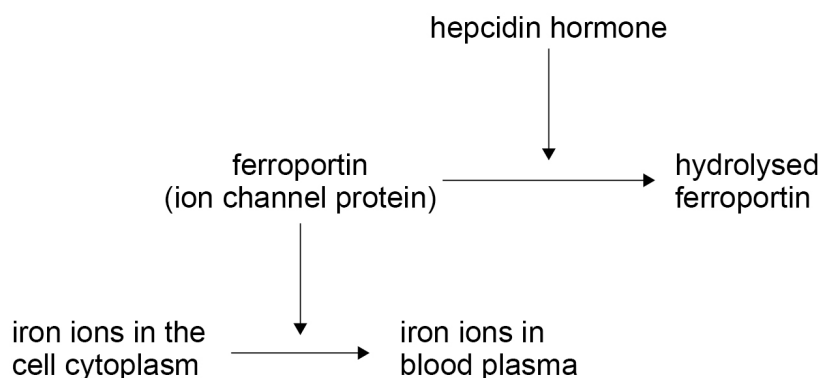
[2 marks]

0 6 . 2

The hormone hepcidin controls the iron ion concentration in blood plasma. Hepcidin affects ferroportin, the iron ion channel protein in cell-surface membranes.

Figure 7 shows how hepcidin controls the iron ion concentration in plasma.

Figure 7



People with the disease haemochromatosis do **not** produce hepcidin.

Use information in **Figure 7** to explain why the iron ion concentration is higher in the plasma of people with haemochromatosis.

[3 marks]

Turn over ►



0 6 . 3

The mass of iron ions in the plasma of a person with haemochromatosis is $6104 \mu\text{g}$
The iron ion concentration in the plasma of a healthy person is $50 \mu\text{g dm}^{-3}$
The volume of blood in each of these people is 4000 cm^3

Calculate the ratio of the mass of iron ions in the plasma of the person with
haemochromatosis to the mass of iron ions in the plasma of the healthy person.

[2 marks]

Answer _____

7